

A Polyhedral-Range Partial Answer to a Residuality Question for Norm-Attaining Operators

FA/Banach discovery run `fa_banach_001`

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Source and Question

This packet concerns:

Mingu Jung, Miguel Martin, Abraham Rueda Zoca, *Residuality in the set of norm attaining operators between Banach spaces*, arXiv:2203.04023.

The source paper studies residuality of norm-attaining operators and density of absolutely strongly exposing operators. Remark 4.2 contains the question addressed here.

where we have used (1) and (2) in the last inequality. This contradicts (3). So, we conclude that $A_{m,k}$ is nowhere dense for each $m, k \in \mathbb{N}$. $\square$

Remark 4.2. It is not possible, in general, to get denseness of Fréchet differentiable points of $\mathcal{L}(X, Y)$ when $\text{NA}(X, Y)$ is residual. For instance, if X has the Radon-Nikodým property, then $\text{ASE}(X, \ell_1)$ is dense in $\mathcal{L}(X, \ell_1)$ (hence, $\text{NA}(X, \ell_1)$ is residual), but there is no point in $\mathcal{L}(X, \ell_1)$ where the norm on $\mathcal{L}(X, \ell_1)$ is Fréchet differentiable since the norm of ℓ_1 is nowhere Fréchet differentiable (use Proposition 1.6). On the other hand, there is no known objection, as far as we know, to get the denseness of $\text{ASE}(X, Y)$ from the residuality of $\text{NA}(X, Y)$ in complete generality. On the other hand, we do not know if the residuality of $\text{NA}(X, Y)$ for a non-trivial Y implies that of $\text{NA}(X, \mathbb{K})$ as the denseness of $\text{ASE}(X, Y)$ does.

Remark 4.3. (1) If the Banach space Y in Theorem 4.1 is *reflexive*, then the denseness of the set of Fréchet differentiable points of $\mathcal{L}(X, Y)$ can be obtained directly from [47, Theorem 3.1]. Indeed, in this case we have that $(X \widehat{\otimes}_{\pi} Y^*)^* = \mathcal{L}(X, Y)$ and the inclusion $\text{NA}(X, Y) \subseteq \text{NA}(X \widehat{\otimes}_{\pi} Y^*, \mathbb{K})$ holds. It follows that $\text{NA}(X \widehat{\otimes}_{\pi} Y^*, \mathbb{K})$ is residual.

The last sentence of the displayed remark asks whether residuality of $\text{NA}(X, Y)$ for a non-trivial range space Y implies residuality of scalar $\text{NA}(X, \mathbb{K})$, as density of $\text{ASE}(X, Y)$ does.

Candidate Partial Result

The theorem below gives an affirmative answer for finite-boundary ranges, including finite-dimensional real polyhedral ranges. It does not address arbitrary ranges.

Theorem 1. *Let X be a Banach space over $\mathbb{K}$. Let $Y \neq \{0\}$ be a Banach space admitting a finite norming boundary modulo rotations: there is a finite set $\Phi \subset S_{Y^*}$ such that*

$$\|y\| = \max_{\phi \in \Phi} |\phi(y)| \quad (y \in Y).$$

Assume that some $\phi_0 \in \Phi$ is non-redundant in the following strong sense: there is $y_0 \in Y$ with $\phi_0(y_0) = 1$ and

$$\beta := \max_{\phi \in \Phi, \phi \neq \phi_0} |\phi(y_0)| < 1.$$

If $\text{NA}(X, Y)$ is residual in $\mathcal{L}(X, Y)$, then $\text{NA}(X, \mathbb{K})$ is residual in X^* .

Corollary 1. *Let Y be a nonzero finite-dimensional real polyhedral Banach space. If $\text{NA}(X, Y)$ is residual in $\mathcal{L}(X, Y)$, then $\text{NA}(X, \mathbb{R})$ is residual in X^* .*

Proof. Let Φ be the set of dual vertices modulo signs. Then

$$\|y\| = \max_{\phi \in \Phi} |\phi(y)|.$$

Choose a vertex ϕ_0 of B_{Y^*} . Since B_{Y^*} is a finite-dimensional polytope, ϕ_0 is exposed: there is $y_0 \in Y$ with $\phi_0(y_0) = 1$ and

$$\max_{\phi \in \Phi, \phi \neq \phi_0} |\phi(y_0)| < 1.$$

The theorem applies. □

A Category Lemma

We use the following standard Baire category fact.

Lemma 1. *Let E and F be Banach spaces, let $U \subset E$ and $V \subset F$ be nonempty open sets, and let $A \subset U$. If $A \times V$ is residual in $U \times V$, then A is residual in U .*

Proof. Equivalently, $(U \setminus A) \times V$ is meagre in $U \times V$. By the Kuratowski–Ulam category theorem applied to the Baire product $U \times V$, the set of $e \in U$ for which the vertical section

$$\{v \in V : (e, v) \in (U \setminus A) \times V\}$$

is meagre in V is residual in U . This vertical section is V when $e \in U \setminus A$ and is empty otherwise. Since V is a Baire space and is not meagre in itself, $U \setminus A$ must be meagre in U . □

Proof of Theorem 1

Put

$$A = \text{NA}(X, \mathbb{K}) \subset X^*.$$

Let $Z = \ker \phi_0$. Since $\phi_0(y_0) = 1$, every $T \in \mathcal{L}(X, Y)$ has a unique decomposition

$$T = y_0 \otimes f + H, \quad f = \phi_0 \circ T \in X^*, \quad H \in \mathcal{L}(X, Z).$$

Thus $\mathcal{L}(X, Y)$ is linearly homeomorphic to

$$X^* \times \mathcal{L}(X, Z).$$

Let

$$C = \max_{\phi \in \Phi, \phi \neq \phi_0} \|\phi|_Z\|.$$

If $\Phi = \{\phi_0\}$, the proof below is interpreted with $C = 0$. Define the open domination region

$$W = \{y_0 \otimes f + H : \|f\| > \beta \|f\| + C \|H\|\}.$$

For $T = y_0 \otimes f + H \in W$, the ϕ_0 -coordinate strictly dominates:

$$\|\phi_0 \circ T\| > \max_{\phi \in \Phi, \phi \neq \phi_0} \|\phi \circ T\|.$$

Indeed, $\phi_0 \circ T = f$, while for $\phi \neq \phi_0$,

$$\|\phi \circ T\| \leq |\phi(y_0)| \|f\| + \|\phi|_Z\| \|H\| \leq \beta \|f\| + C \|H\|.$$

We claim that on W ,

$$T \in \text{NA}(X, Y) \iff \phi_0 \circ T \in \text{NA}(X, \mathbb{K}). \quad (1)$$

If $\phi_0 \circ T$ attains its norm at $x \in S_X$, then

$$\|Tx\| \geq |\phi_0(Tx)| = \|\phi_0 \circ T\| = \|T\|,$$

so T attains its norm. Conversely, suppose T attains its norm at $x \in S_X$. Since Φ is a finite norming boundary, there is $\phi \in \Phi$ such that

$$|\phi(Tx)| = \|Tx\| = \|T\|.$$

Hence $\|\phi \circ T\| = \|T\|$. The strict domination on W forces $\phi = \phi_0$, and therefore

$$|\phi_0(Tx)| = \|\phi_0 \circ T\|.$$

So $\phi_0 \circ T$ attains its norm.

Now assume $\text{NA}(X, Y)$ is residual in $\mathcal{L}(X, Y)$. Let

$$U_m = \{f \in X^* : \|f\| > 1/m\} \quad (m \in \mathbb{N}).$$

Choose $r_m > 0$ such that

$$\beta \|f\| + C \|H\| < \|f\| \quad (f \in U_m, \|H\| < r_m).$$

For instance, if $C > 0$ one may take

$$r_m = \frac{1 - \beta}{2Cm}.$$

Then

$$U_m \times B_{\mathcal{L}(X, Z)}(0, r_m) \subset W.$$

By residuality of $\text{NA}(X, Y)$, relation (1), and the product coordinates,

$$A \times B_{\mathcal{L}(X, Z)}(0, r_m)$$

is residual in

$$U_m \times B_{\mathcal{L}(X, Z)}(0, r_m).$$

Lemma 1 gives that A is residual in U_m . This holds for every m . Since

$$X^* \setminus \{0\} = \bigcup_{m=1}^{\infty} U_m$$

and $0 \in \text{NA}(X, \mathbb{K})$, the set $A = \text{NA}(X, \mathbb{K})$ is residual in X^* .

Why This Avoids the Rank-One Slice Obstruction

A previous route tried to restrict a residual subset of $\mathcal{L}(X, Y)$ to a closed rank-one subspace. That is not valid in general: a residual set in a Banach space can have poor intersection with a closed nowhere-open subspace.

The proof here instead uses open product regions

$$U_m \times B_{\mathcal{L}(X, Z)}(0, r_m).$$

On those open sets, finite-boundary domination makes norm-attainment in Y equivalent to scalar norm-attainment. Baire category is stable under restriction to open sets and reflects through the full product with the second factor, which is the key difference.

Literature and Novelty Check

Local source inspection showed that arXiv:2203.04023 proves several forward directions: scalar residuality or density of strongly exposing functionals implies range residuality/density for many classes of ranges, including polyhedral compact-operator situations. The reverse implication above for finite-boundary ranges is not stated there.

Bounded web searches on June 13, 2026 used exact and close phrases including

"residuality of NA(X,Y)" "NA(X, K)"

and phrases involving norm-attaining operators, residuality, polyhedral range, scalar residuality, finite boundary, and the title of arXiv:2203.04023. These searches found the source paper and adjacent norm-attainment papers, but no separate explicit answer to this finite-boundary reverse direction.

Limitations and Review Focus

The result does not solve the full question in Remark 4.2. It relies on an isolated boundary coordinate. Smooth finite-dimensional ranges, especially the Euclidean plane, remain outside the argument.

Human review should check:

- the domination equivalence (1);
- the Baire product reflection lemma in the possibly nonseparable Banach setting;
- the specialization from finite-boundary ranges to real finite-dimensional polyhedral ranges.

No computational verification is needed.